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Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 4: System Level Requirements

Version: 1.00

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1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	11:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 System Level Requirements

The Airbag ECU is a safety-critical embedded system responsible for detecting crash events and commanding airbag deployment within the automotive environment. The hardware must reliably process sensor signals, make deployment decisions, and communicate fault states all within the tight timing constraints of a real-world crash event (typically less than 30 ms from impact to deployment).

4.1 Timing Requirements Operating Environment

Tabelle 2 Timing Requirements Operating Environment

Parameter	Specification
Temperature Range	–40 °C to +85 °C (automotive grade)
Mechanical Shock	Per LV124 standard
EMI Requirements	ECE R10 compliant
Supply Voltage	9–16 V DC (nominal 12 V)
Post-Crash Power	Capacitor-backed ($\geq$ 150 ms hold-up)
Communication Bus	CAN 2.0B, ID 0x120, 8 bytes
Deployment Timing	Typically, less than 30 ms from impact to deployment

4.2 Safety-Relevant Functions ASIL-D Compliance Mandates

Tabelle 3 Safety-Relevant Functions ASIL-D Compliance Mandates

ID	Function	Reference
F1	Crash detection via acceleration, gyroscope, and pressure sensors	Airbag_ECU_Phase_2, 4.1 Identified Hazards, Table 2, H1, H2
F2	Airbag squib firing (simulated via digital output in prototype)	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG2
F3	Sensor monitoring and plausibility checking	Airbag_ECU_Phase_3, 4.2 Technical Safety Mechanisms, Table 3
F4	System self-test and diagnostics	Airbag_ECU_Phase_3, 4.2 Technical Safety Mechanisms, Table 3
F5	Fail-safe / fall-back on detected hardware faults	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG1
F6	CAN communication of safety-relevant states to vehicle ECU	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG1, SG2

4.3 Fault Tolerance Requirements Safe State Trigger Conditions

The hardware safe state is defined as: Driver airbag output = LOW, Passenger airbag output = LOW, Warning lamp = HIGH, CAN fault frame transmitted.

Tabelle 4 Safe State Trigger Conditions

ID	Trigger Condition	Hardware Implication
FS01	ADC value out of valid range (less than 2 or greater than 1020)	Sensor open/short circuit; hardware fault block deployment
FS02	Output readback mismatch (commanded HIGH, reads LOW)	Output driver failure; hardware fault activates warning lamp
FS03	CAN bus error detected (bus-off state)	Communication hardware fault enters safe state
FS04	Baby seat signal intermediate voltage (neither 0 V nor 3.3 V)	Seat switch hardware fault passenger airbag suppressed, warn
FS05	Watchdog timer expiry (MCU stall)	Hardware watchdog resets MCU; outputs go to default LOW (safe)

4.4 Compliance Considerations

Tabelle 5 Compliance Considerations

Standard / Requirement	Reference
ISO 26262 Functional Safety	Airbag_ECU_Phase_2, 4.3 ASIL Classification Matrix, Table 4
ASIL-D requirements for airbag system	Airbag_ECU_Phase_2, 4.3 ASIL Classification Matrix, Table 4
Automotive EMC and reliability standards	Airbag_ECU_Phase_5, 4.6 Key Design Principles, Table 6

5 Figure and table listing

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6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
3	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

No. Reference

- [1] International Organization for Standardization (ISO), ISO 26262: Functional Safety Standard for Modern Road Vehicles. https://fscdn.rohm.com/en/products/databook/white_paper/iso26262_wpe.pdf
- [2] S. N. Easwaran, M. Mollat, D. Sreedharan et al., Fault Tolerant Low Side Driver Circuit Design Using Design FMEA for Automotive Applications, IEEE NEWCAS, 2019. doi: 10.1109/newcas44328.2019.8961225
- [3] G. Bahig and A. El-Kadi, Formal Verification of Automotive Design in Compliance With ISO 26262, IEEE Access, vol. 5, pp. 4505–4516, 2017. doi: 10.1109/access.2017.2683508
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8 Version history

Document name	date	Change log
Airbag_ECU_Phase_4	09.06.2026	v1.00 Initial version. System Level Requirements compiled from HW Concept Design Report V02 Sections 2, 9 and Architecture V2 Section 8.